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Transforming Cognitive Maps into Markov Process Models for Software Development Projects

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***Abstract*—The complexity of managing software development projects in organizational and technical systems is due to a number of features. Such projects are weakly structured systems. The parameters of weakly structured systems form a complex "web" of connections and states that change over time. The development of projects in such a multifactorial environment can only be represented in the form of qualitative models. The use of cognitive maps with their subsequent presentation in the form of Markov chains allows quantifying the course of project processes, which are the main condition for the success of the project. At the same time, quantitative estimates represent not only a multi-vector picture of the state of projects, but also have the property of forecasting. The scope of the proposed method can be significantly expanded through the use in the educational process in the training of professionals in the field of programming and project management.**

Keywords—IT-project; cognitive map; Markov's chain; project-oriented systems; SWEBOK; directed graph; matrix of transition

I. INTRODUCTION

In modern society, the rate of economic growth and the level of development of countries is largely determine by the role of scientific and technological progress in the intellectualization of any sphere of our life. Organizations that provide favorable conditions for effective innovation win in global competition. Innovative projects and programs are implement in the form of major intersectoral projects to create, master and disseminate new information technologies that contribute to fundamental changes in many areas of science and technology.

A competent specialist is distinguishing by critical thinking, the ability to choose the optimal one among a variety of solutions, and reasonably refute false solutions. Competence implies the constant updating of knowledge and skills, the possession of new information for the successful solution of professional tasks in an environment where the external environment is constantly changing [1]. In other words, "competence is the ability to actually perform activities" both independently and as part of the project team [2].

The basis of such activities is the preparation and adoption of effective decisions in project management based on acceptable goal-setting methods for the proactive management of projects for the creation and development of systems, products and technologies [3]. At the same time, due to the multifactorial nature of project management systems, one should take into account the possibility of compromise, balancing and harmonizing solutions, and this is possible only through the use of modern modeling methods, which allows not only to obtain a result by changing the parameters and states of the system, but also forms a general vision of the system in dynamics, and as well as understanding the features of the interaction of technical, behavioral and contextual competencies in the course of project work [4].

Management of project-oriented systems of any profile presupposes the presence of subsystems for analysis, preparation, and effective decision-making. The creation of such information technologies for project management is impossible without the development of models, methods and mechanisms for the interaction of the project team.

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Orientation towards the creation of such information technologies leads to the improvement of methods and models. Changes in approaches to the development of information technologies lie in the fact that it is necessary to move from frozen schemes to modeling activities in the constantly changing conditions of society and production [5]. Modeling complex situations based on cognitive analysis allows us to resolve this contradiction.

The initial concept in cognitive modeling of complex processes is the concept of a cognitive map, which is a directed weighted graph, in which [6]:

- the vertices correspond to the basic factors (states) of the project, which can be verified using data mining technology, which makes it possible to discard redundant factors that are weakly related to the core of the basic factors;
- direct links between factors reflect causal chains along which the influence of a certain factor extends to other factors, and it is believed that the factors included in the “if ..., then ...” condition affect the factors of the consequence of the entire chain, and this effect can be either reinforcing (positive), or inhibitory (negative), or of variable sign, depending on possible additional conditions.

The cognitive map displays only the structure of relationships between factors. It does not reflect the essence of the impact, as well as the dynamics of changes in influences depending on changes in the situation or changes in time factors. The display of these features is possible at the next level of information structuring in a cognitive model [7]. At this level, any connection between factors can be reveal to the corresponding equation, which can contain both quantitative (measured) variables and qualitative (fuzzy) expressions. In this case, quantitative variables enter naturally in the form of their numerical values. Each qualitative variable is associated with a set of linguistic variables that reflect different states of this variable (for example, the functionality of a software product can be “low”, “satisfactory”, “higher than the requirements of the terms of reference for the project”), and each linguistic variable corresponds to a certain numerical equivalent according to scale [0,1], for example, using the Harrington desirability function [8]. With the accumulation of knowledge about the processes, it becomes possible to reveal in more detail the nature of the relationships between factors.

An important step in cognitive modeling is the construction of a cognitive map, which is a directed graph, the vertices of which are related factors, and the arcs are causal relationships between factors [9]. In this case, the "+" sign means a positive connection, and "-" corresponds to a negative connection [10].

II. Main part

The life cycle of a software product includes development, deployment, support, and maintenance [11]. In most cases, any complex software product, as a rule, is implement by separate independent projects. Maintenance includes the elimination of critical faults in the system and is implement as a process activity. Support consists of developing new functionality, redesigning existing functionality due to changing requirements, and improving the product. The life cycle of a software product ends with the decommissioning of the product and its removal from support and maintenance.

Planning and implementation of IT projects is carried out considering the specific requirements, the characteristics of the knowledge and competencies of the performers [12].

The complexity of managing software development projects in organizational and technical systems is due to several features:

- the presence of many factors and their interconnectedness;
- lack of sufficient information about the dynamics of processes;
- turbulence of the environment and variability of the nature of the processes in time.

Such projects are loosely structured systems. Many system factors form a complex system of connections and states that change over time. The development of projects in such a multifactorial system cannot be described using deterministic models. Decision-making procedures for project management can be based on cognitive analysis and modeling random processes [13].

Consider the construction of a cognitive map on the example of software development project management.

The most common approach to software development at present is versioning, in which the stages of development and debugging of the program code are sequentially performed, and the evaluation of the result is reduced to the "as it turns out" formula. This approach, as a rule, provides software development at acceptable cost and quality, but this process includes a lot of random errors and trials, which are "team knowledge" and based on specific performers [14].

According to SWEBOK [15], software development includes 10 main and 7 additional knowledge areas (KA’s) on which software development processes are based:

- Software requirements;
- Software design;
- Software construction;
- Software testing;
- Software maintenance;
- Software configuration management;
- Software engineering management;
- Software engineering process;
- Software engineering tools and methods;
- Software quality/

Additional KA’s include:

- Computer engineering;
- Computer science;
- Management;
- Mathematics;
- Project management;
- Quality management;
- Systems engineering.

All this is necessary to know and be able to apply to develop software. Project management in the field of software development is one of the 17 KA’s, and then auxiliary.

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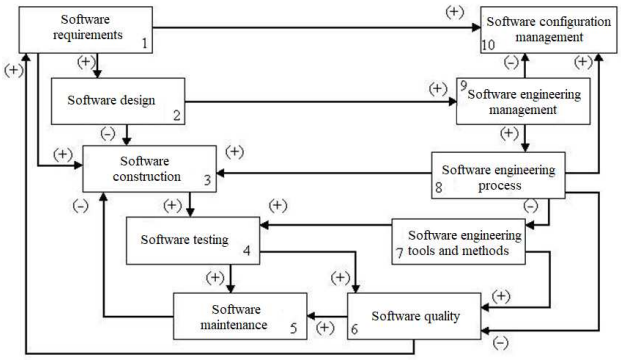

Figure 1. Software Development Cognitive Map

The cognitive map of software development includes 10 vertices corresponding to the main KA's, competencies, and connections between these vertices (Fig. 1). In fact, such a cognitive map reflects the states of the system and the transitions between these states. If we accept that the sum of the probabilities of all states is equal to one, and that the transition from each state to another is incompatible events, then such a graph can be represent as a homogeneous Markov chain with discrete states and discrete time [16].

To do this, we supplement the directed graph, which reflects the cognitive features of software development projects, with links between each of the 10 processes (states) and get a mark chain.

III. Model Development and Use of Modelling Method

The indicated transformation of the cognitive map into a Markov chain allows one to move from qualitative assessments of the progress of projects to quantitative characteristics. At the same time, quantitative estimates are not only a multi-vector picture of the state of projects, but also have the property of forecasting.

The matrix of conditional transition probabilities for this Markov chain will look as shown in Fig.2.

The value of transition probabilities π_{ij} was obtained using expert methods.

$$\|\pi_{ij}\| = \begin{Vmatrix} \pi_{1,1} & \pi_{1,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \pi_{1,10} \\ 0 & \pi_{2,2} & \pi_{2,3} & 0 & 0 & 0 & 0 & 0 & \pi_{2,9} & 0 \\ 0 & 0 & \pi_{3,3} & \pi_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \pi_{4,4} & \pi_{4,5} & \pi_{4,6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \pi_{5,3} & 0 & \pi_{5,5} & 0 & 0 & 0 & 0 & 0 \\ \pi_{6,1} & 0 & 0 & 0 & \pi_{6,5} & \pi_{6,6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \pi_{7,4} & 0 & 0 & \pi_{7,7} & 0 & 0 & 0 \\ 0 & 0 & \pi_{8,3} & 0 & 0 & \pi_{8,6} & \pi_{8,7} & \pi_{8,8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \pi_{9,8} & \pi_{9,9} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \pi_{10,9} & \pi_{10,10} \end{Vmatrix} = \begin{Vmatrix} 0{,}4 & 0{,}4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0{,}2 \\ 0 & 0{,}5 & 0{,}3 & 0 & 0 & 0 & 0 & 0 & 0{,}2 & 0 \\ 0 & 0 & 0{,}2 & 0{,}8 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0{,}4 & 0{,}4 & 0{,}2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0{,}6 & 0 & 0{,}4 & 0 & 0 & 0 & 0 & 0 \\ 0{,}1 & 0 & 0 & 0 & 0{,}2 & 0{,}7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0{,}3 & 0 & 0 & 0{,}7 & 0 & 0 & 0 \\ 0 & 0 & 0{,}1 & 0 & 0 & 0{,}15 & 0{,}3 & 0{,}45 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0{,}2 & 0{,}8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0{,}6 & 0{,}4 \end{Vmatrix}$$

Figure 2. Transition probability matrix for the original Markov chain

Based on the matrix of transition probabilities, provided that the initial state of the system is known, it is possible to determine the probabilities of the states p1(k), p2(k),…, p10(k) after any k-th step (1). On fig. 3 shows the results of modeling the states of the system for the initial matrix of transition probabilities.

The resulting probabilities of states allow predicting and evaluating the effectiveness of the project. It should be noted that as the project progresses, the degree of resource intensity of individual processes changes.

$$p_i(k) = \sum_{j=1}^{m} [p_j(k-1) \cdot \pi_{ji}]\Big|_{m=6}; \quad i = 1, 2, \cdots, n \tag{1}$$

IV. Conclusions

For the considered basic software development project and the classic IT project team, as well as based on the totality of transition probabilities that were determined by the expert method, the following conclusions can be drawn. The highest state probability for the conditions π3.4 = 0.4 and π3.3 = 0.6 after the 10th step corresponds to process 3 – "Software construction" (Fig. 3). Further, processes 4 and 5 are the most resource intensive. "Software quality" can also be attributed to the most important processes – curve 6.

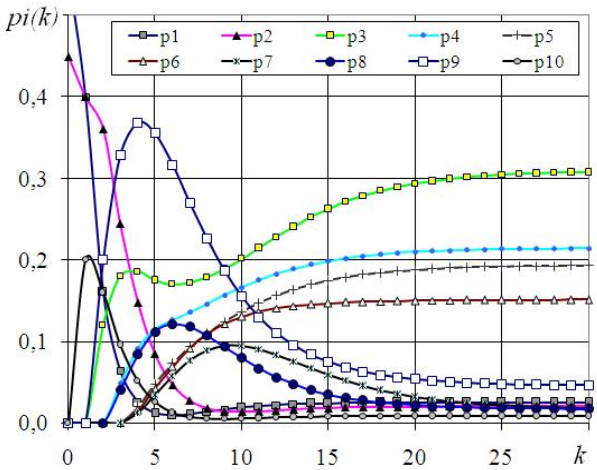

Figure 3. Change in the probabilities of the states of the processes for the conditions π3.4 = 0.4 and π3.3 = 0.6

For the values of state probabilities π3.4 = 0.4 and π3.3 = 0.7 after the 10-th step, the picture of the project performance changes significantly. The main time-consuming processes are processes 4 "Software Testing" and 5 "Software Maintenance". Processes 3 "Software construction" and 6 "Software Quality" move to the third and fourth positions, respectively (Fig. 4).

The main cause of the most software project failures is the use of inadequate development management practices. Classical control methods stop working in cases where the structure and properties of the managed object are unknown to us and/or change over time. These approaches will also not help if the current properties of the object do not allow it to develop in accordance with the required characteristics. If the project team cannot provide the necessary efficiency and constantly works in a rush mode, then this does not lead to an increase in productivity, but to the departure of professionals from the project and its failure.

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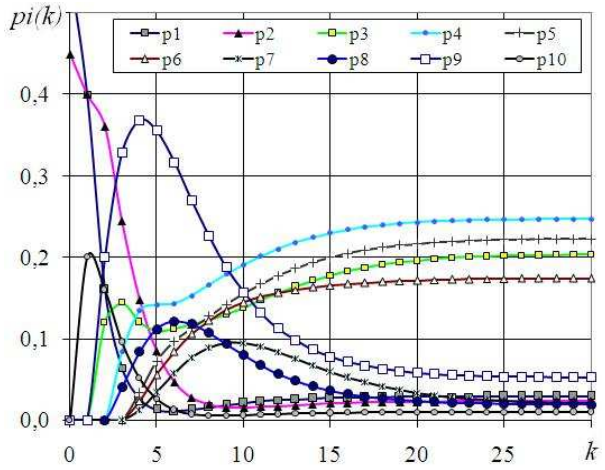

Figure 4. Changing the probability of process states for conditions $\pi 3.4 = 0.3$ and $\pi 3.3 = 0.7$

The transformation of cognitive maps into Markov chains makes it possible to quantify the course of project processes, which is an essential condition for the success of projects. The area of application of the proposed method can be significantly expand through the use in the educational process for competence-based training in the preparation of professionals in the field of programming and project management.

REFERENCES

[1] V. N.Burkov, D. A.Novikov, *How to manage organizations. M: Sinteg, 2014.*

[2] D. Lukianov, K. Bespanskaya-Paulenka, V. Gogunskii, O. Kolesnikov, A. Moskaliuk, and K. Dmitrenko, "Development of the markov model of a project as a system of role communications in a team", *EEJET, vol. 3 (87),* pp. 21–28, Jun. 2017.

[3] S. Bushuev, N. Bushueva, "The mechanisms of the formation of values in the activities of design-driven organizations", *EEJET, vol. 1-2 (43), pp. 4–9*, April 2010

[4] A. Stanovsky, K. Kolesnikova, E. Lebedeva, and I. Kheblov, "Dynamic models in the method of project management", *EEJET, vol. 6, iss. 3(78),* pp. 46–52, Dec. 2015.

[5] R.M. Bell, J.S. Jones, S.R. Murray, "Using Leadership Values to Minimize Resistance and Facilitate Change: An Analysis of 1 Peter 5", *in Henson, J.D. (eds) Biblical Organizational Spirituality. Christian Faith Perspectives in Leadership and Business.* Palgrave Macmillan, Cham, 2022, pp.419 – 448. https://doi.org/10.1007/978-3-031-04006-1_21

[6] V. Beglytsia, S. Tymofeev, A. Gozhyj, "Cognitive analysis and modeling of the investment situation in Ukraine", *in CEUR Workshop Proceedings*, vol. 2762, 2020, pp. 235-243.

[7] V. C. Prakash, J. K. R. Sastry, G. Reeshmika, M. Pavani, P. C. Sree, J. S. R. Teja, "Development of a comprehensive and integrated expert system for career assessment based on cognitive models" *in International Journal of Emerging Trends in Engineering Research*, vol. 7(11), 2019, pp. 617-627. doi:10.30534/ijeter/2019/347112019

[8] K. Kolesnikova, T. Olekh, D. Lukianov and D. J. Obenewaa, "Markov Principles of Project Effectiveness", *Proceedings of the 2021 IEEE International Conference on Smart Information Systems and Technologies*, pp. 1-6, 2021. doi: 10.1109/SIST50301.2021.9465881.

[9] E. Tolman. Cognitive maps for rats and humans [online] Available: http://sknowledge.ru/Msg.aspx?id=11807

[10] Cognitive Technologies for Management Decision Support. [online] Available: http://emag.iis.ru/arc/infosoc/emag.nsf/BPA/092aa276c601a997c32568c0003ab839

[11] Abadeh, M. N. (2020). Performance-driven software development: An incremental refinement approach for high-quality requirement engineering. Requirements Engineering, 25(1), 95-113. doi:10.1007/s00766-019-00309-w

[12] Y. P. Bakara, T. Raharjo, B. Hardian, A. Suhanto, "Project quality management challenges and solutions in IT project: A systematic literature review", *Proceedings of the 2021 IEEE International Conference on Computing, Engineering and Design, ICCED* 2021, pp. 1-6. doi:10.1109/ICCED53389.2021.9664861

[13] A.E. Karlik, E. A. Iakovleva, V. V. Platonov, "Semantic modelling for the communication mechanism of the production system", *Knowledge in the Information Society*, *vol. 184,* pp. 355-366, November 2020. DOI: 10.1007/978-3-030-65857-1_30

[14] A. Kolomiiets, O. Mezentseva, K. Kolesnikova, "Customer churn prediction in the software by subscription models it business using machine learning methods" *in CEUR Workshop Proceedings,* vol.3039, 2021, pp. 119-128.

[15] Software Engineering Body of Knowledge (SWEBOK). [online] Available: https://www.computer.org/education/bodies-of-knowledge/software-engineering

[16] V. Piterska, O. Kolesnikov, D. Lukianov and other, "Development of the markovian model for the life cycle of a project's benefits", *EEJET, vol.5, iss. 4(95),* pp. 30 – 39, Dec. 2018 doi: 10.15587/1729-4061.2018.145252

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